

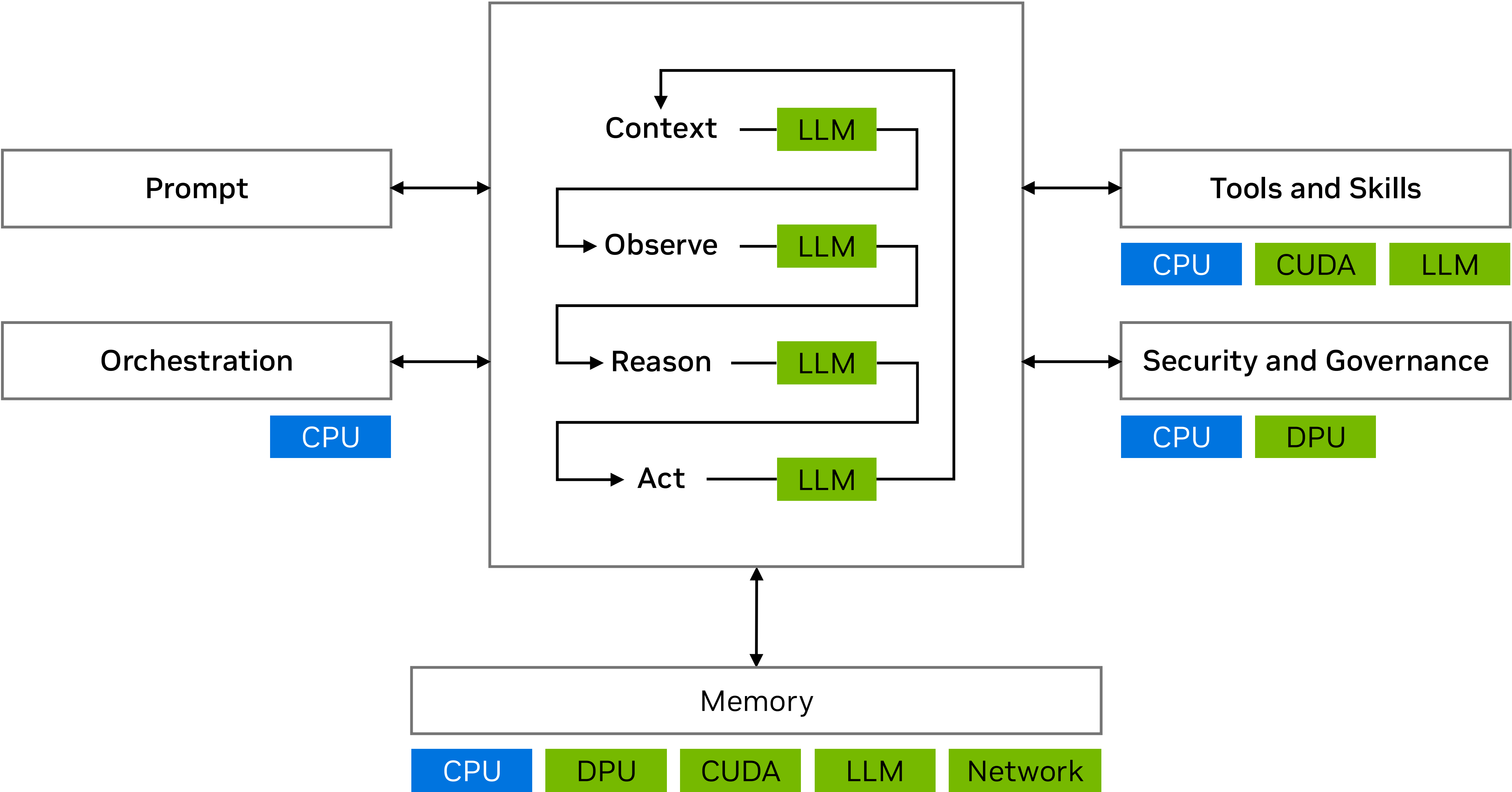


NVIDIA Vera CPU

Jonathon Evans, Polychronis Xekalakis
Hot Chips 2026



Agentic AI is the Most Complex Computing Workload in History



Beyond the GPU - NVIDIA Vera Rubin is a Full-Stack AI Factory Platform

Extreme co-design for agentic AI across seven chips and five racks

Vera Rubin
NVL72



Foundation Platform

Groq 3
LPX



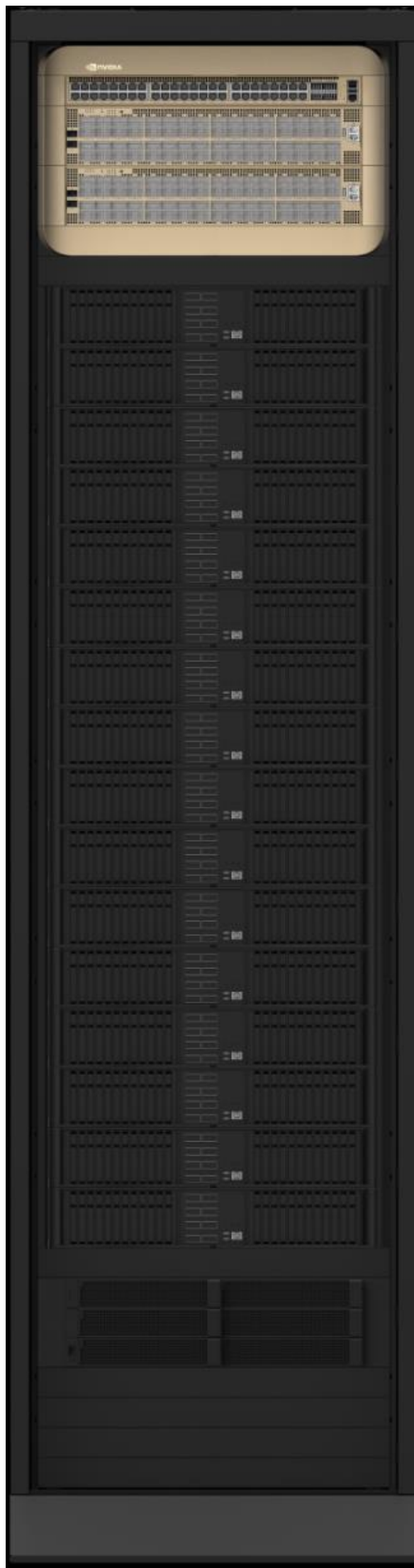
Extend Interactivity

Vera CPU
Rack



Tool Calls and Sandboxes

Vera BlueField-4
STX Storage



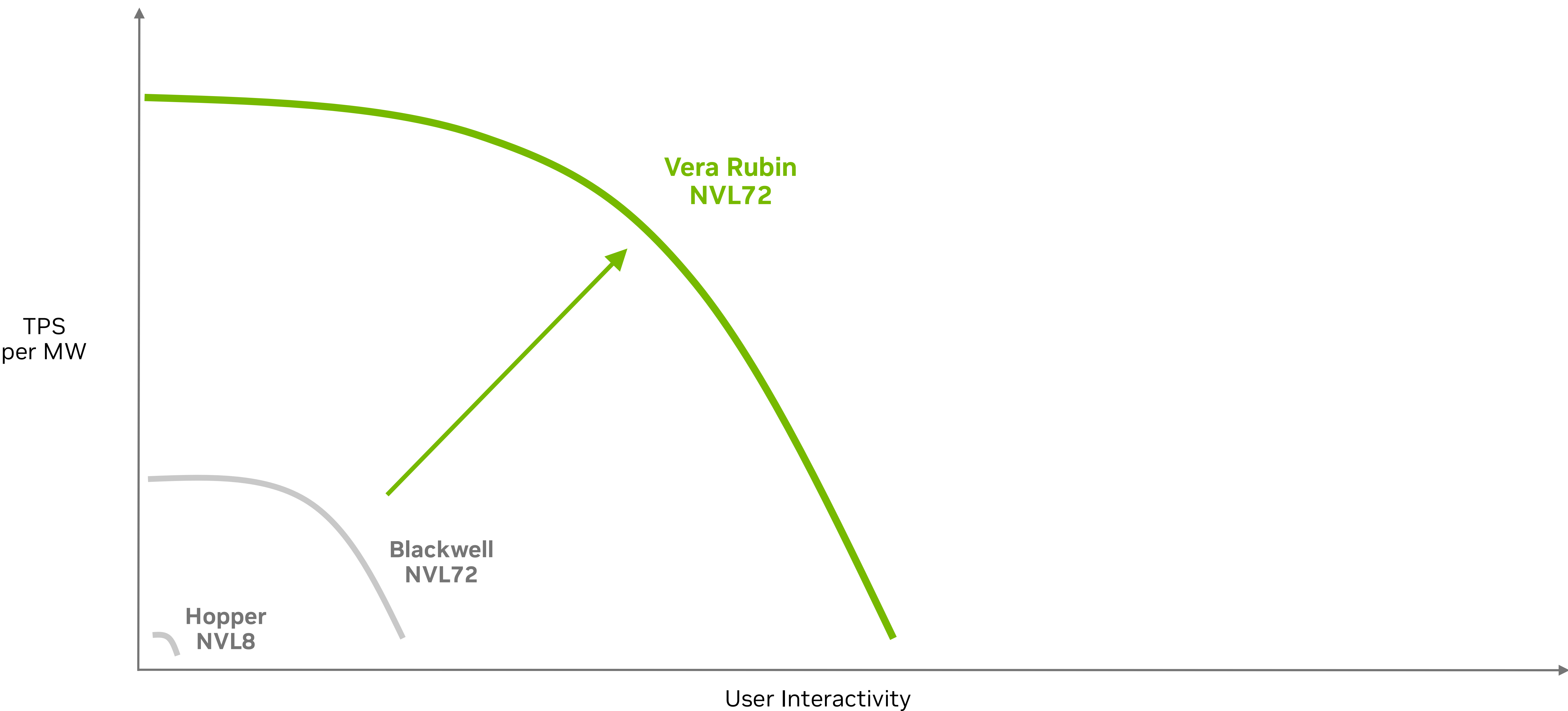
Context Memory Storage

Spectrum-6 SPX
Networking



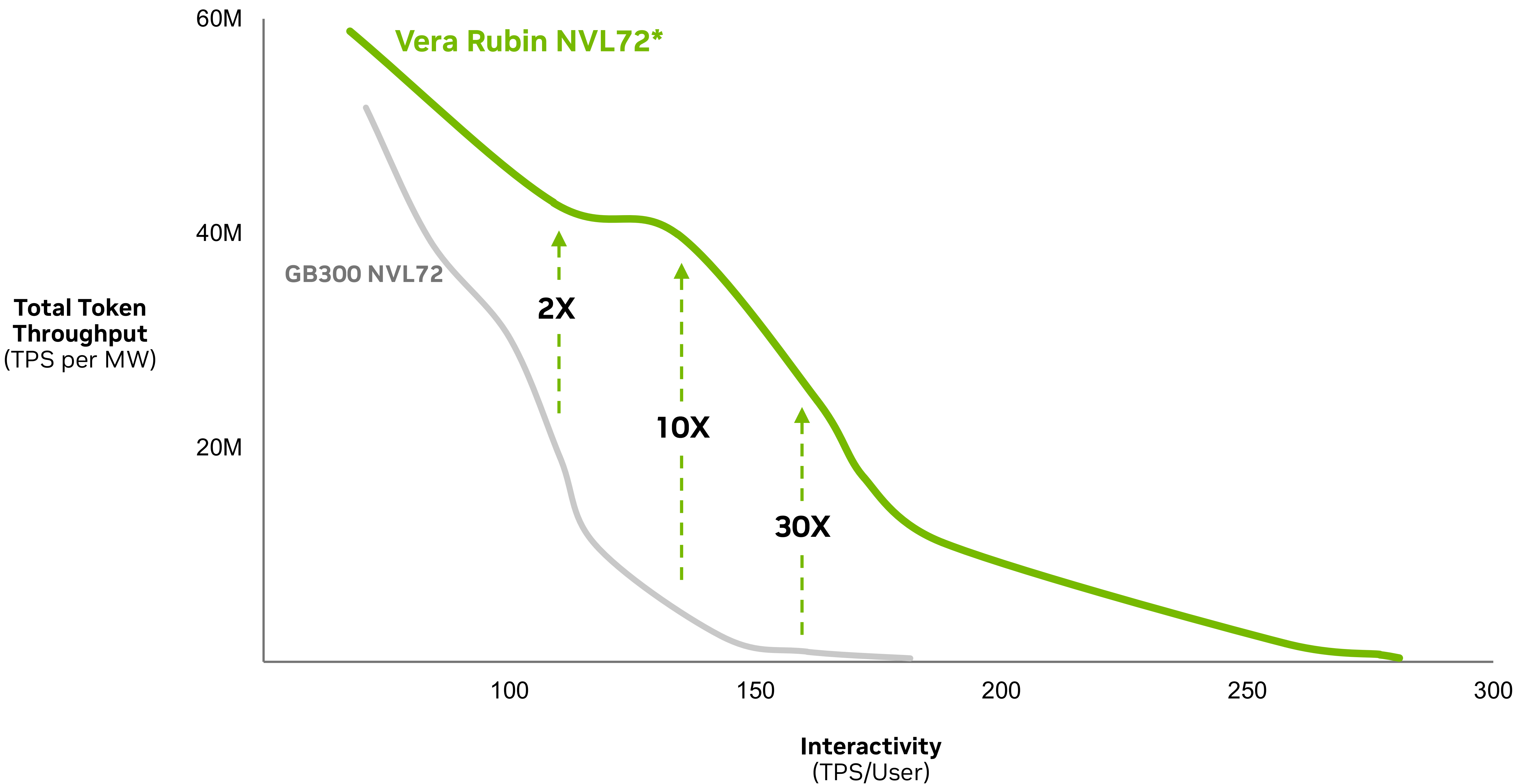
Scale-Out Fabric

Vera Rubin NVL72 is the Foundation of Every AI Factory



Vera Rubin NVL72 Delivers Up to 30X Throughput Per MW

DeepSeek-v4-PRO | 140K+ Context | SemiAnalysis AgentX

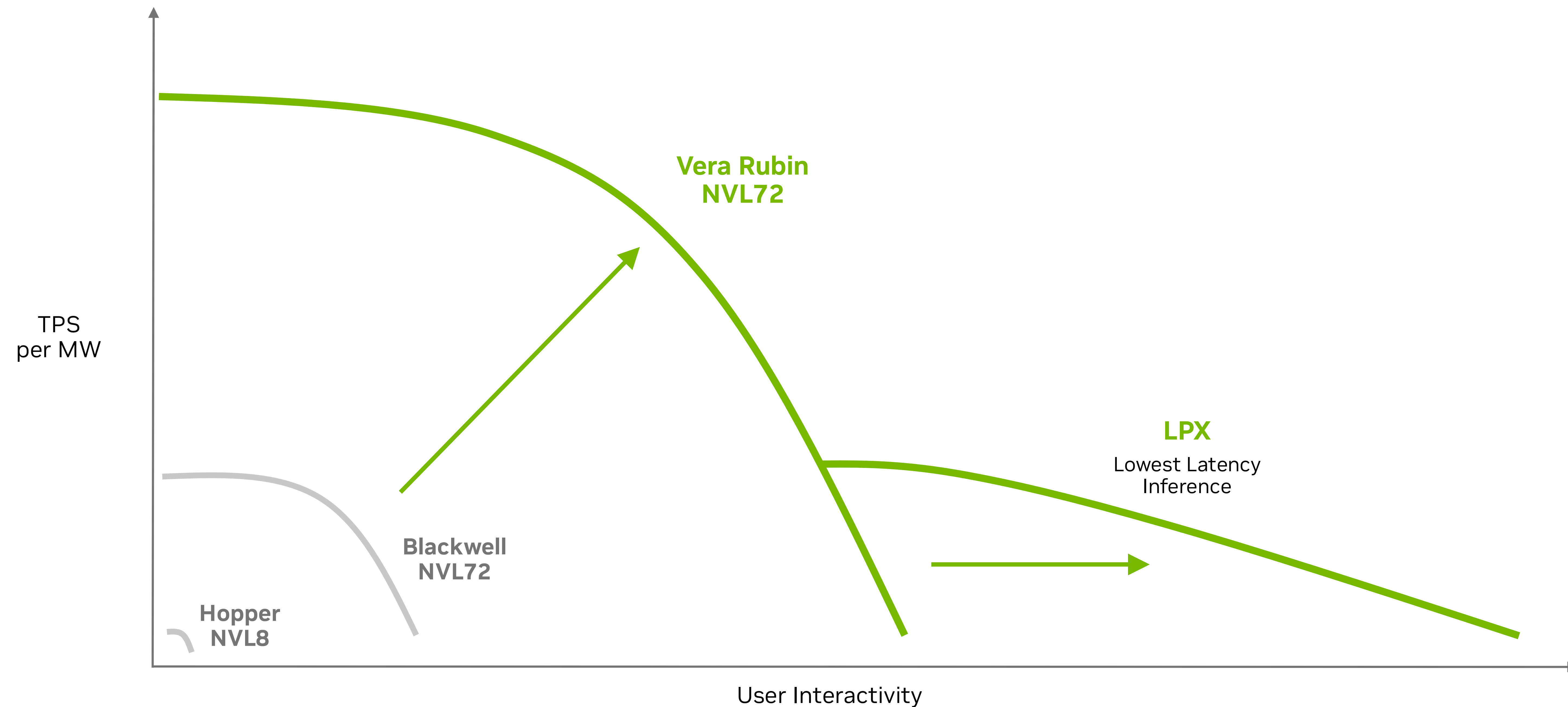


*Unofficial Results. Pending Semi-Analysis Review



NVIDIA Groq 3 LPX — Sets New Standard for AI Interactivity

Improving user experiences for agentic workloads



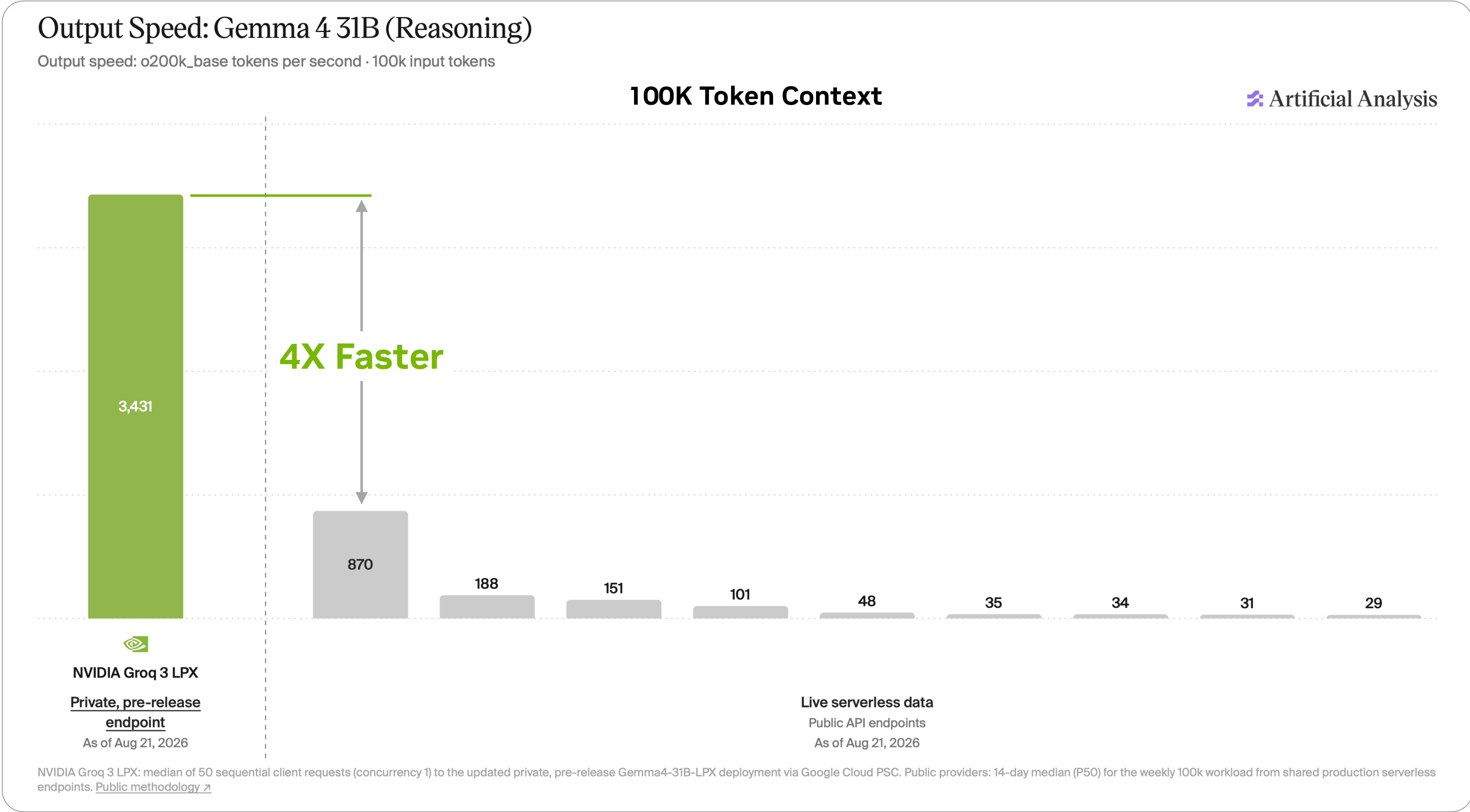
LPX 3rd Party Benchmark — 4X Faster Long-Context Decode

Record measured by Artificial Analysis with 100K context length for agents

NVIDIA Groq 3 LPX



In Full
Production

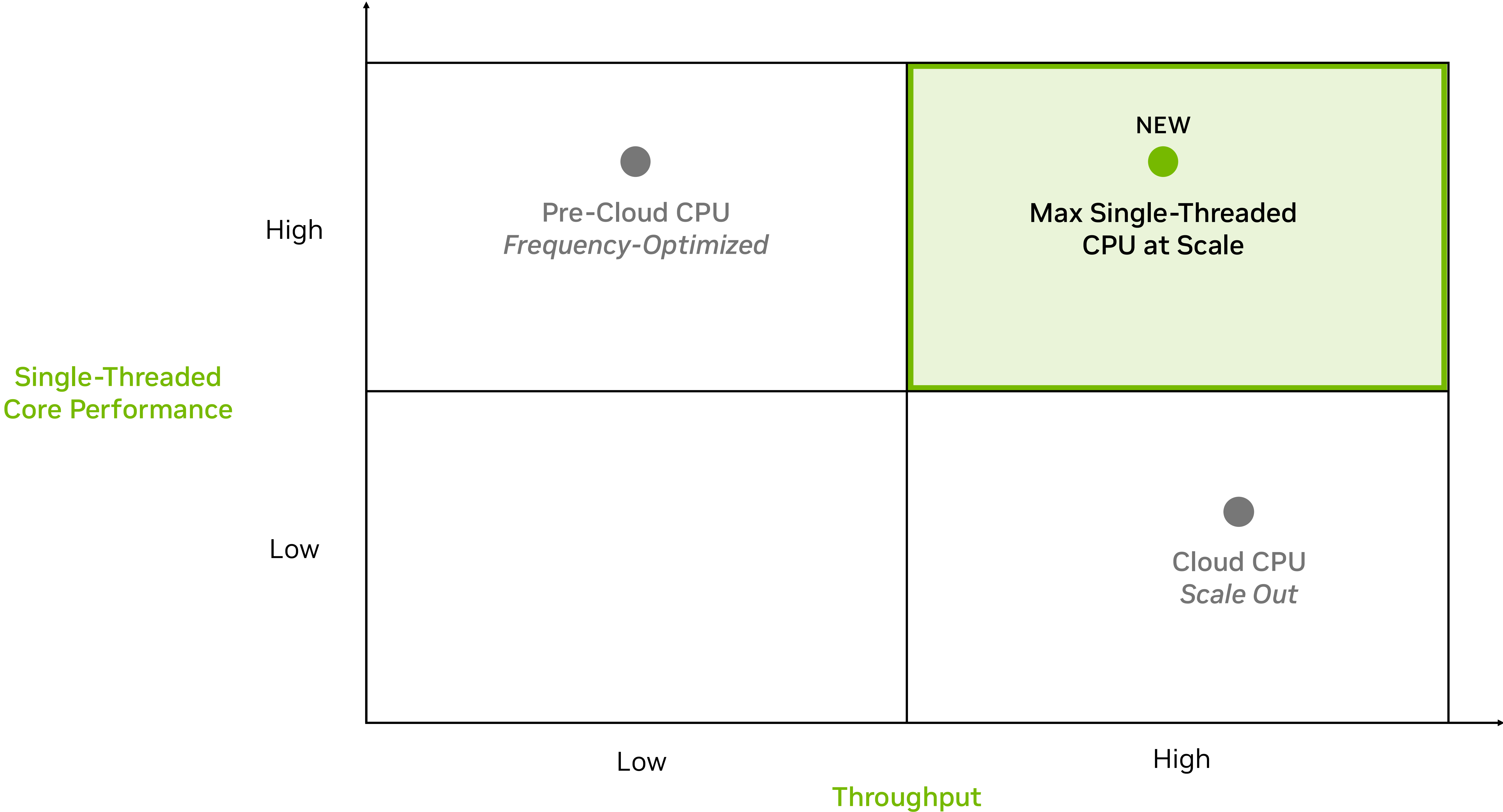


Vera CPU



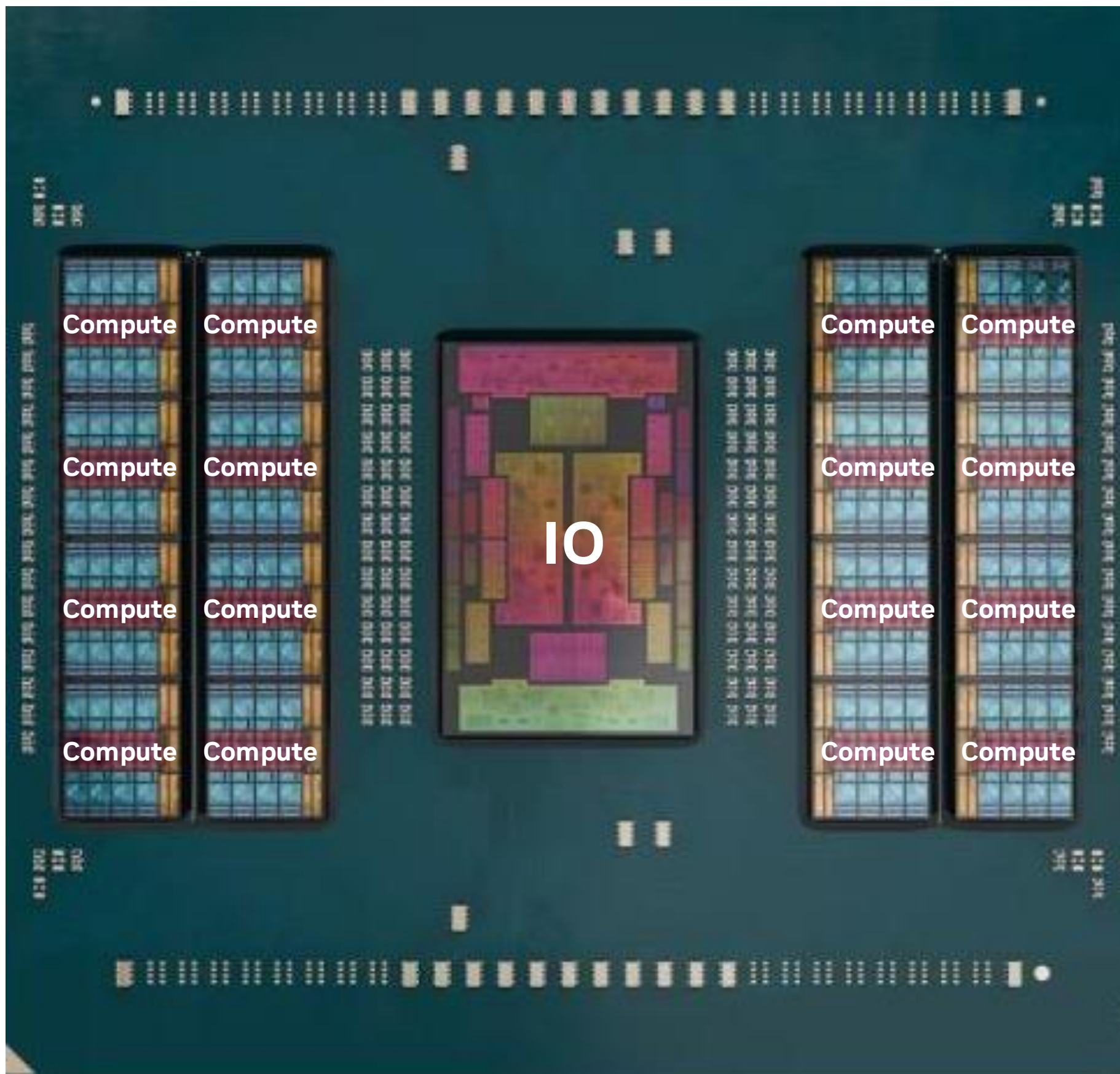
Vera CPU Maximizes Agent Performance per Core – Not Core Count

The max single-threaded CPU at scale

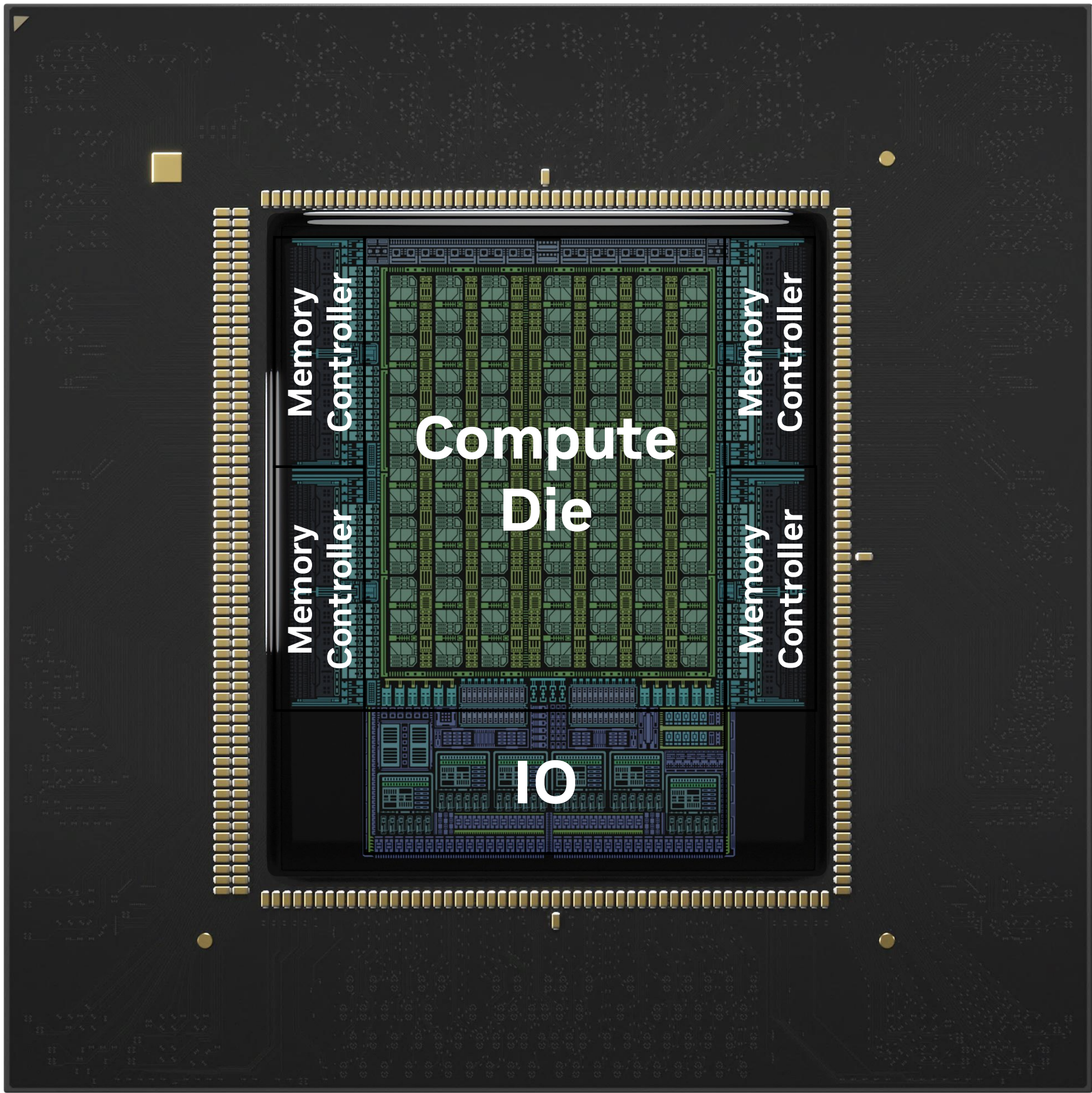


Vera CPU: Low-Latency Compute That Won't Keep AI Agents Waiting

Exemplifies the max single-threaded CPU at scale



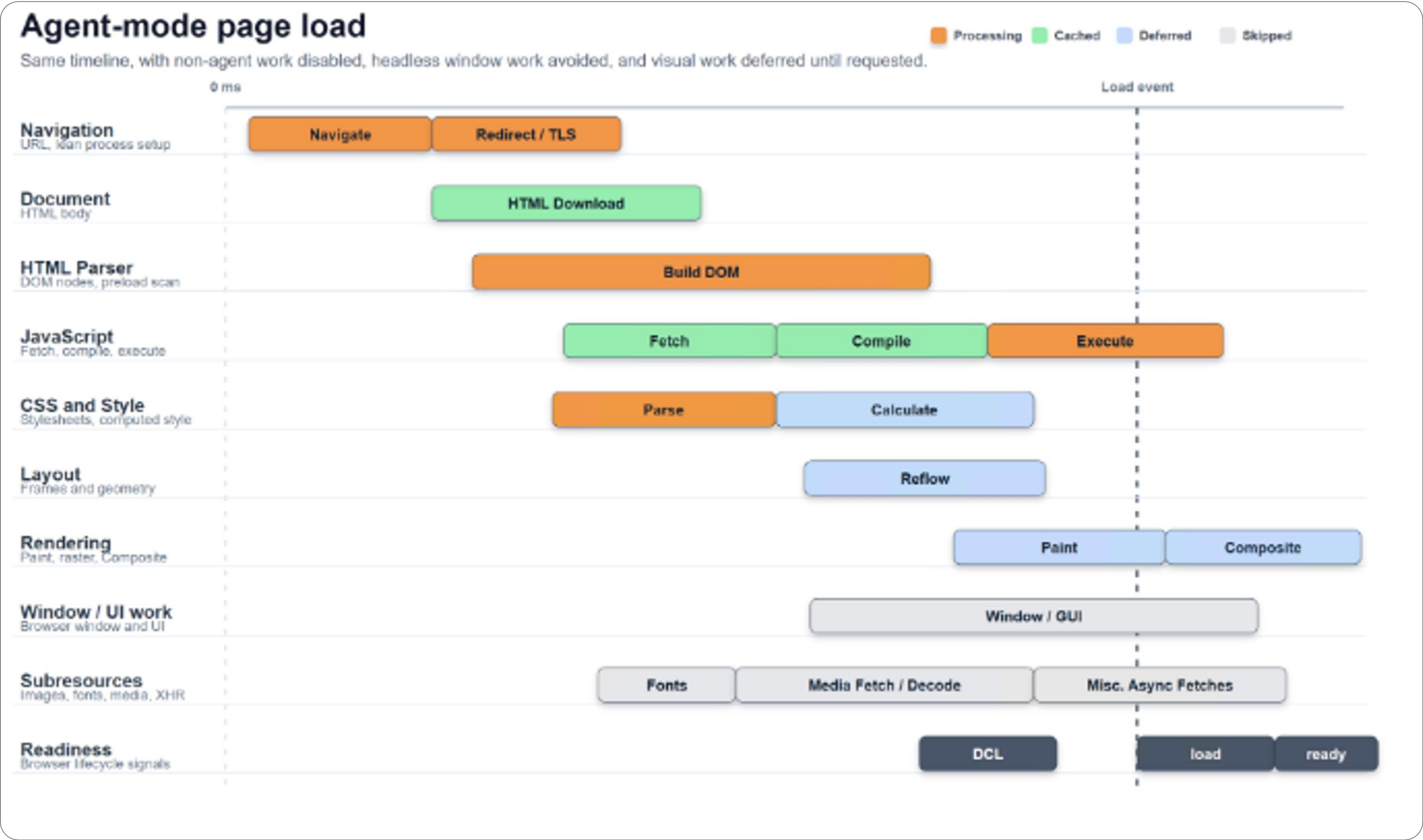
Traditional Chiplet Architecture



NVIDIA Vera CPU

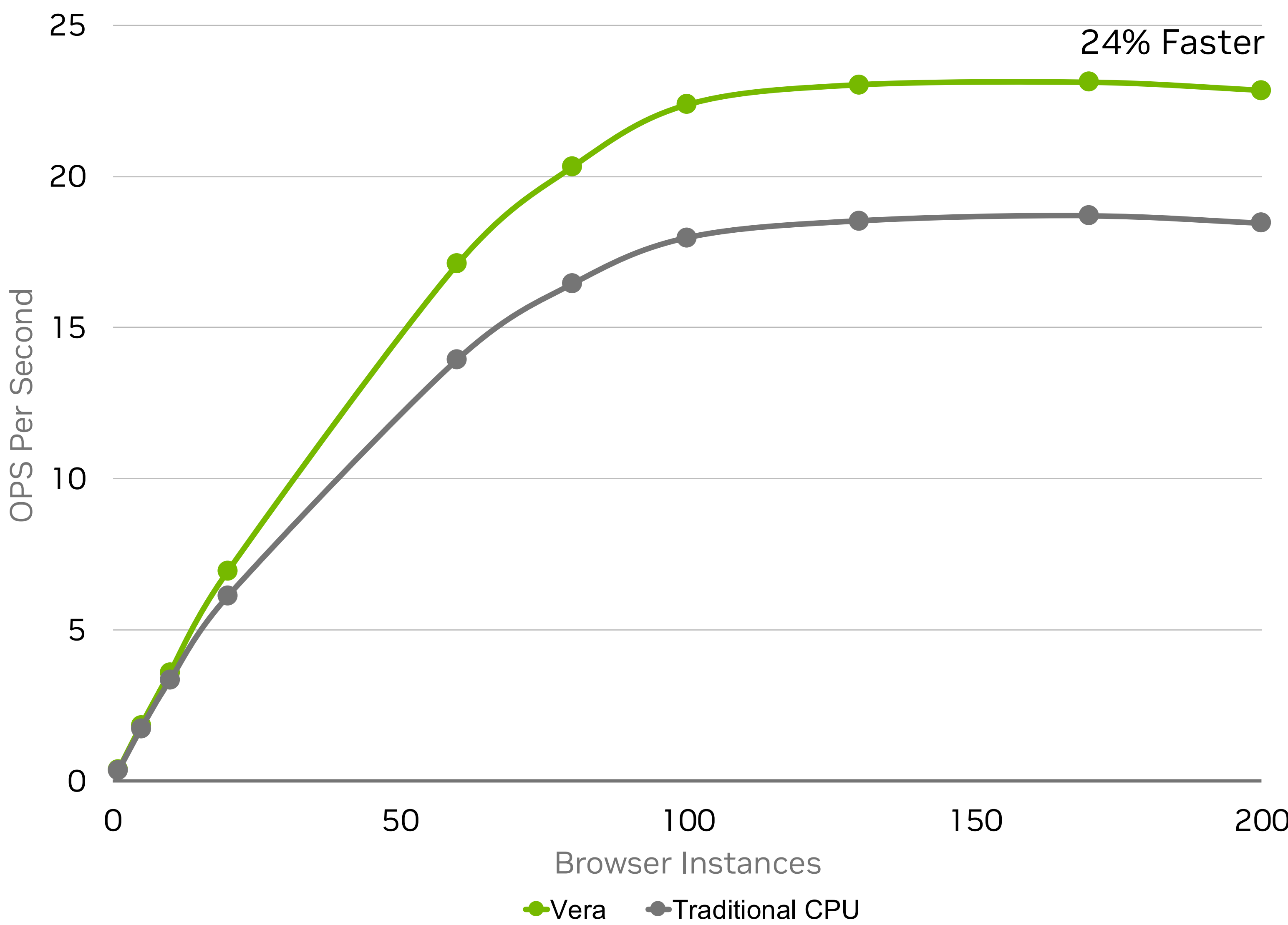
Vera Drives Massive Speedups Across Real-World Agentic Workflows

Headless browsers



4.5X Removing the rendering part and caching re-use, browsers can be sped up by up to 4.5X

Speedometer With Chromium

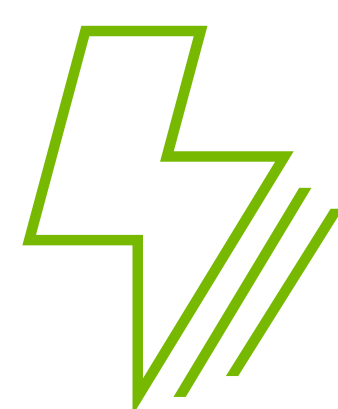


Five Innovations Differentiate Vera From Conventional Data Center CPUs

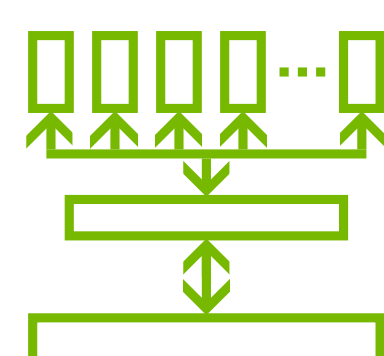
Agents need CPUs with



Massive
Cores



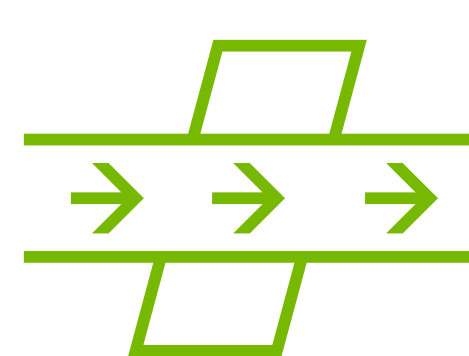
Built for Power
Limited Data Centers



Built to Process
Massive Data



Completely
Protected

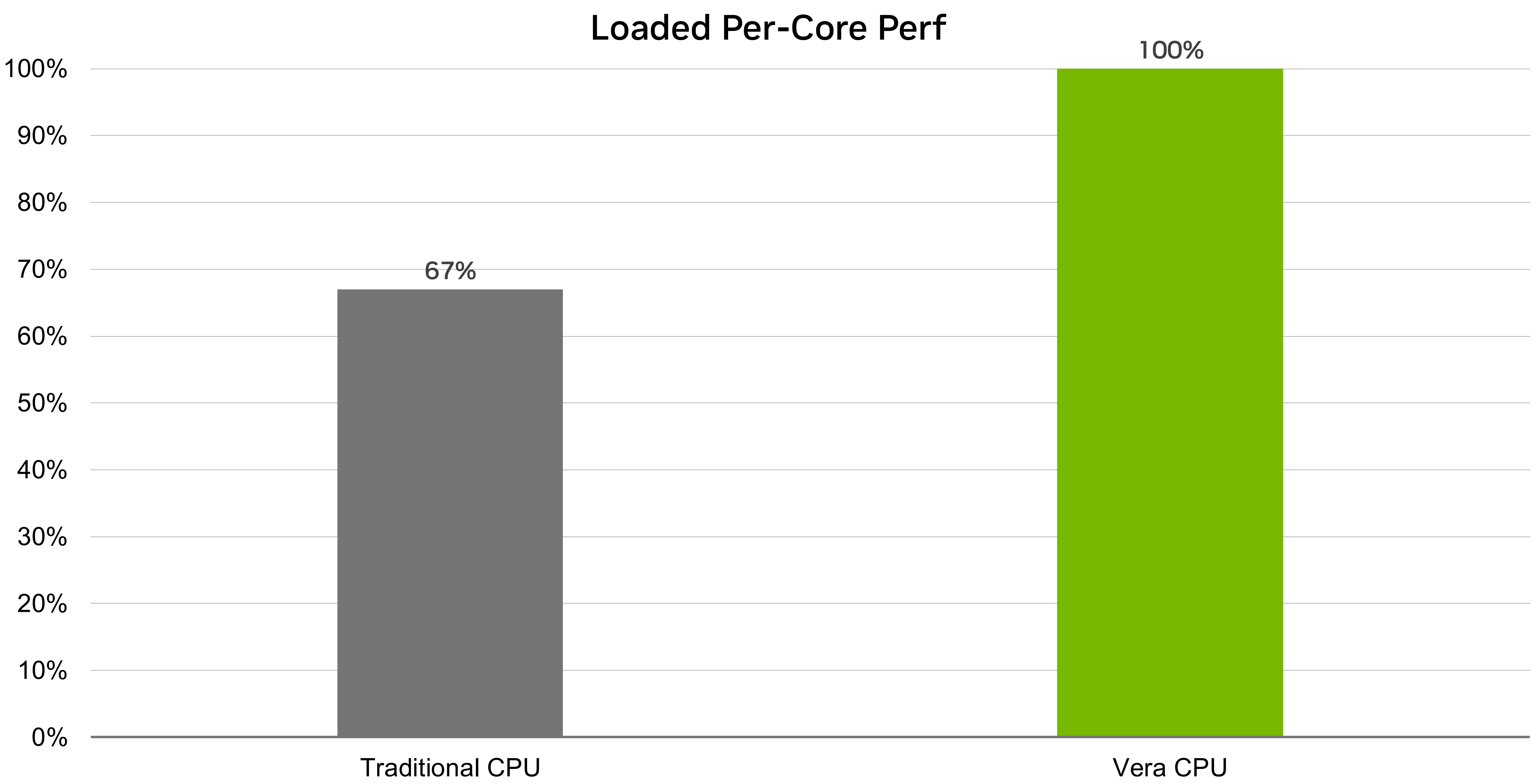


Massive I/O

Differentiator 1: Massive Cores

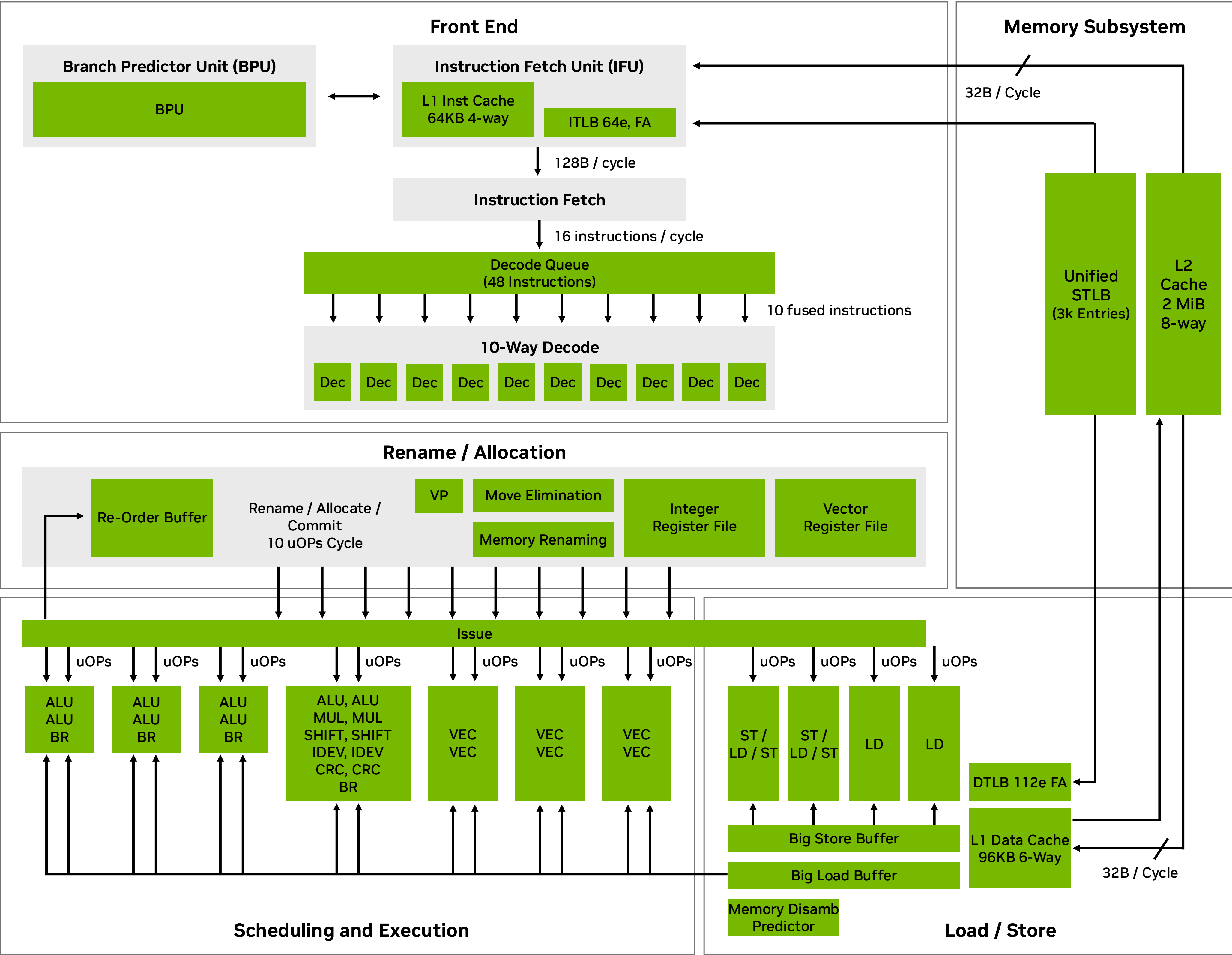


Massive Cores Deliver the Deterministic Single-Thread Performance AI Agents Demand



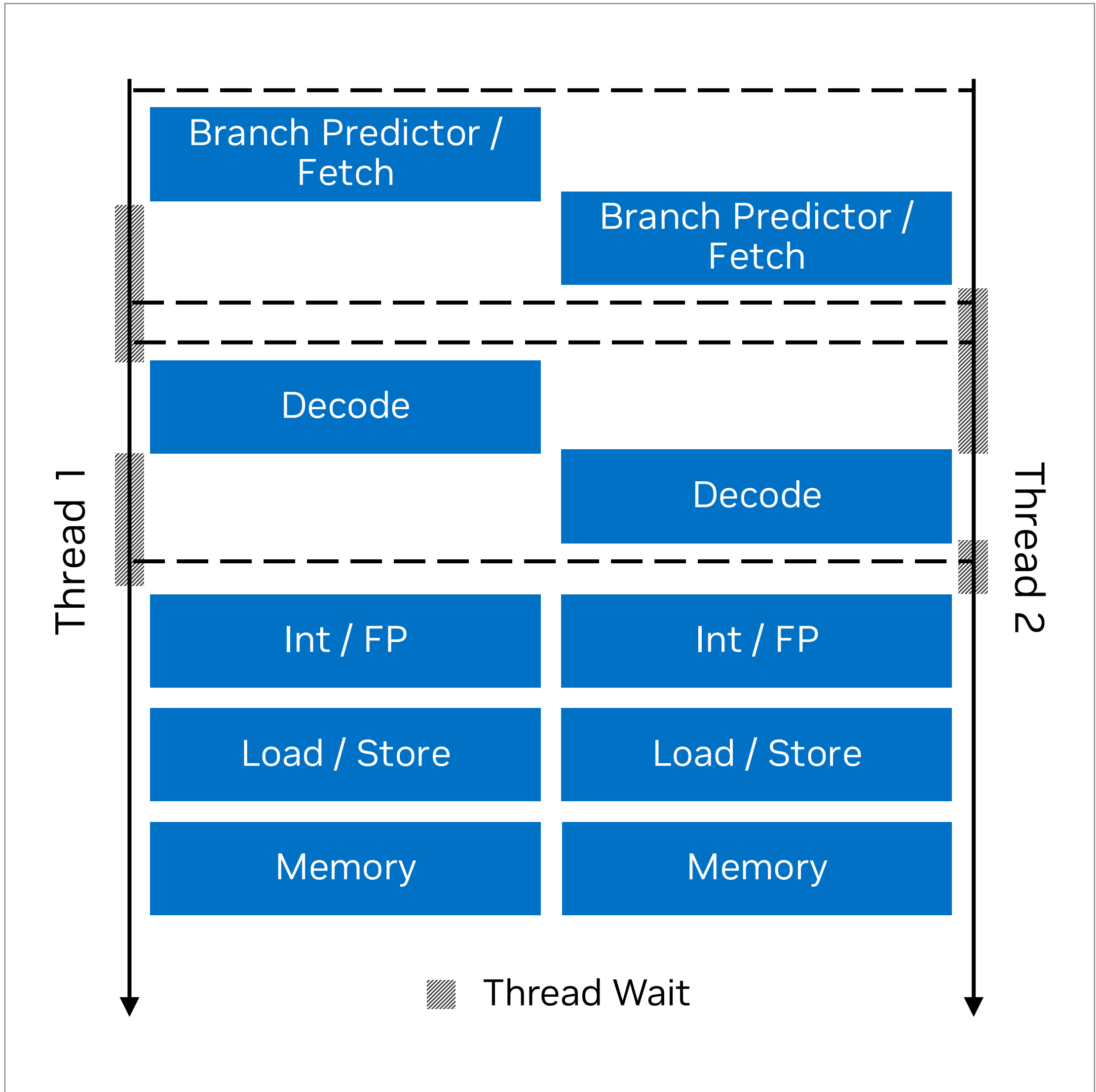
Olympus Delivers More Agent Performance per Core, per Watt

High performance, area and energy efficient custom cores

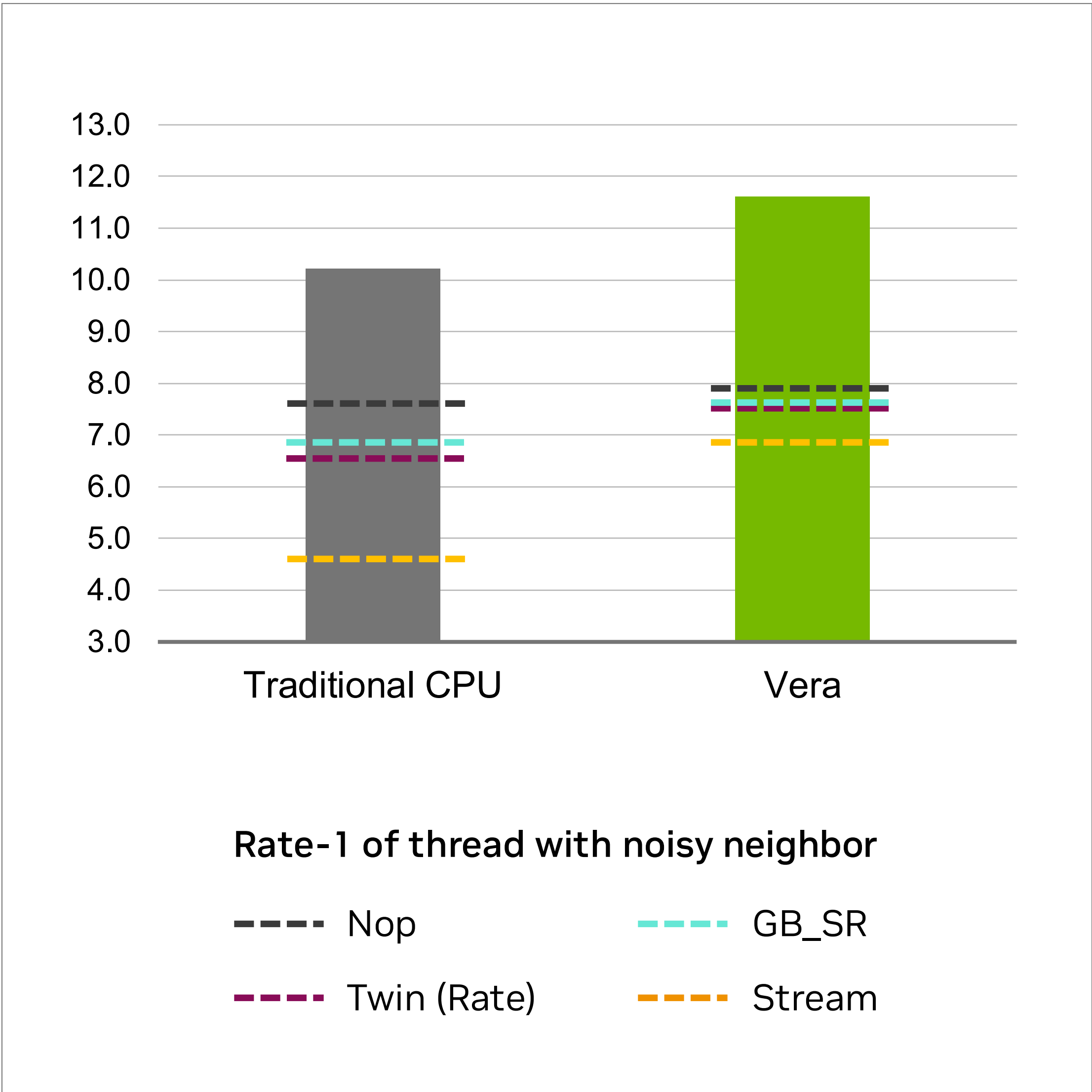


Spatial Multithreading Makes High Performance Deterministic

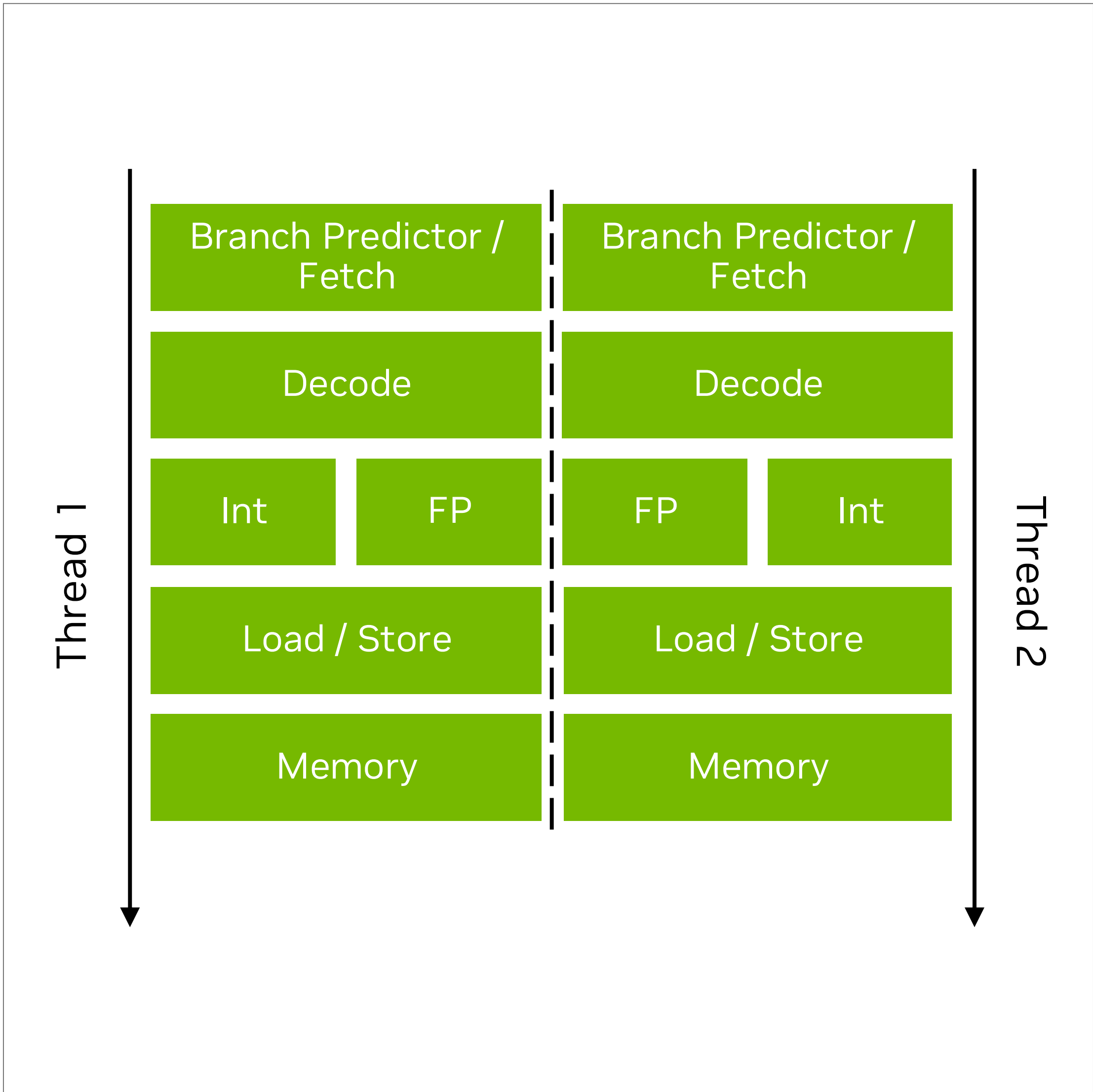
Deterministic high performance



Traditional SMT (x86)
Noisy neighbor and resource contention



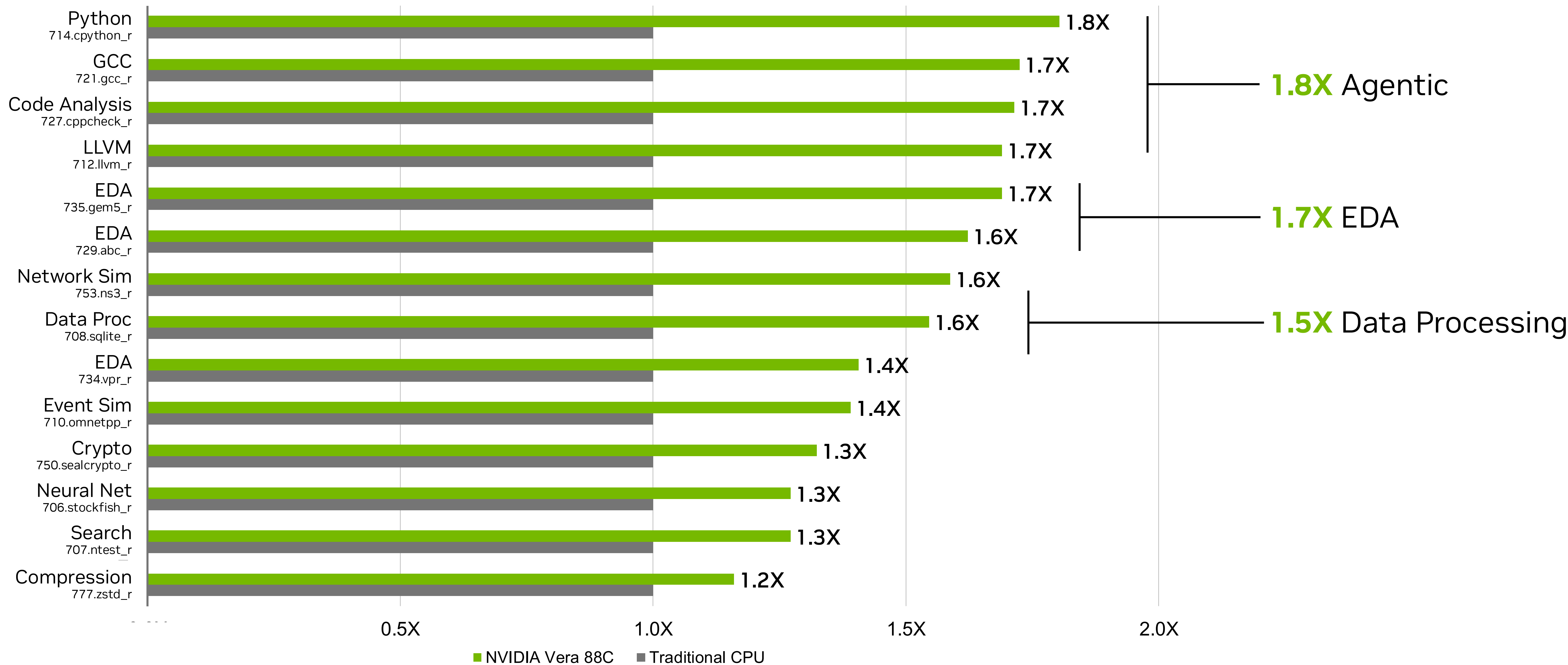
Per Core Score SpecInt2k17 Rate-1



NVIDIA Spatial Multi-Threading
Determinism and isolation

Vera and Olympus — A Single Thread Perf Machine

SPEC CPU® leadership



Comparison based on relative performance for individual SPECrate®2026_int_base benchmark ratios between an NVIDIA Vera CPU and an AMD EPYC 9755 measured internally July 2026. Selected benchmarks shown are those most relevant to agentic AI workloads. Results are estimated due to the use of a pre-production reference system but otherwise follow all SPEC CPU 2026 Run and Reporting rules. SPEC®, SPEC CPU®, and SPECrate® are registered trademarks of the Standard Performance Evaluation Corporation (www.spec.org).



Differentiator 2: Built for Power Limited Datacenter

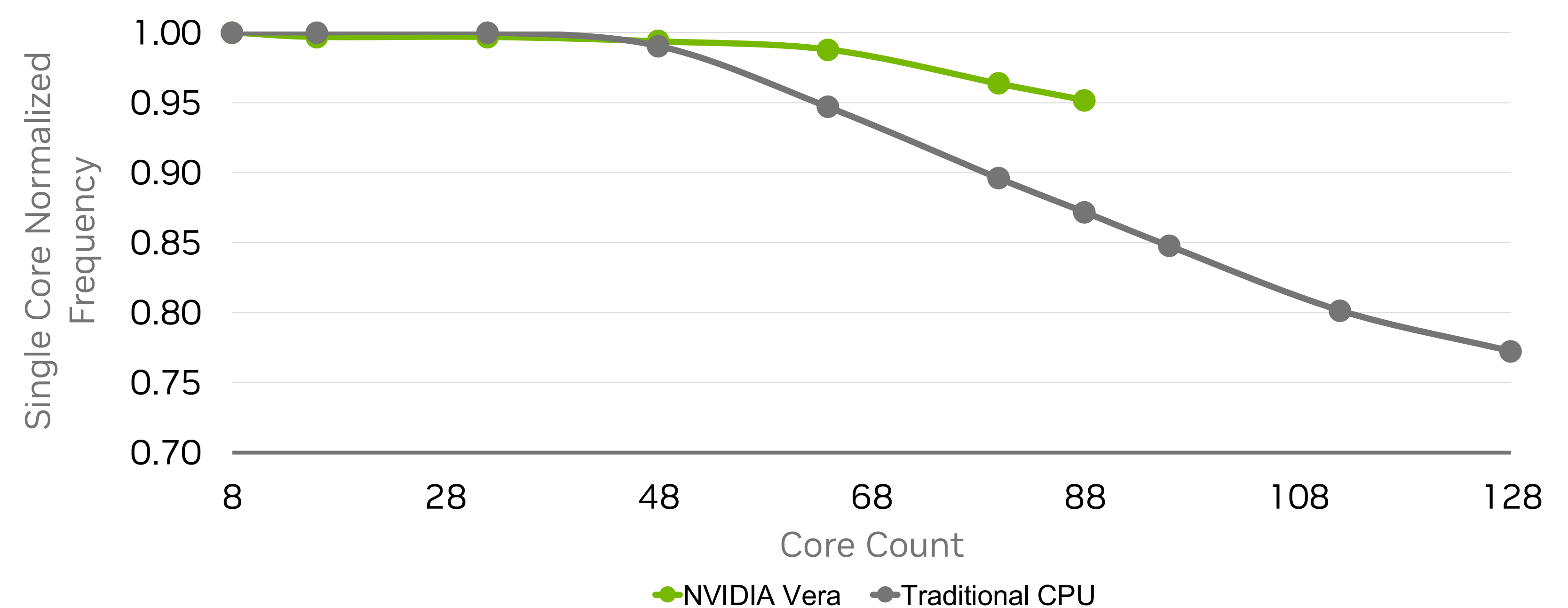
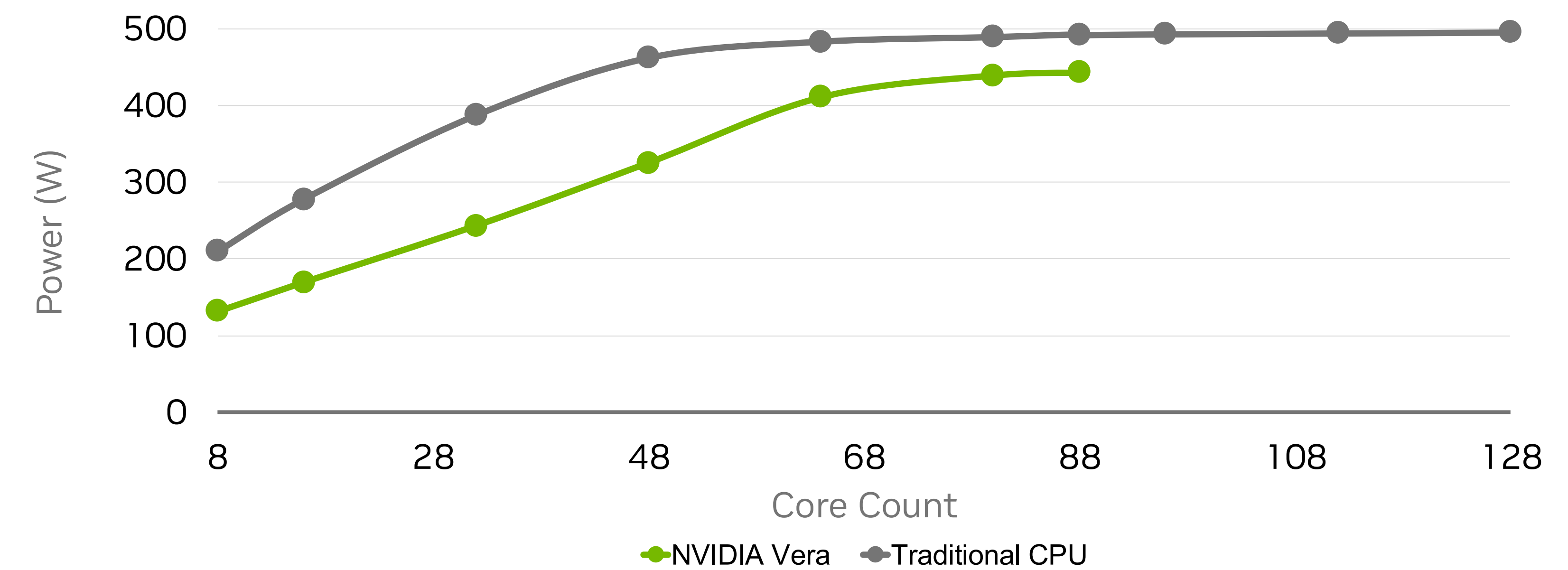


Power and Bandwidth Provisioning is Critical for Agents

Performance-under-load is key

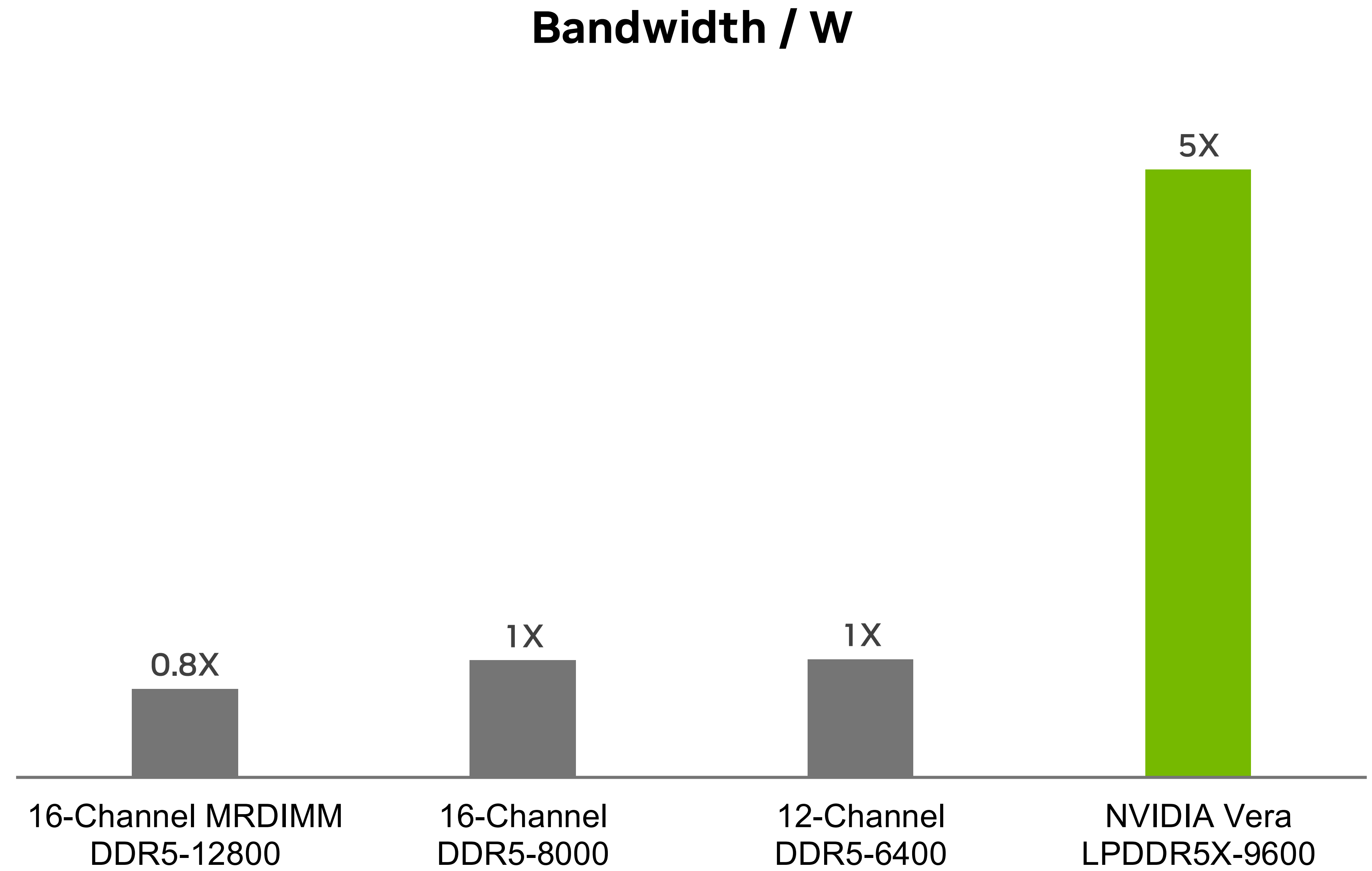
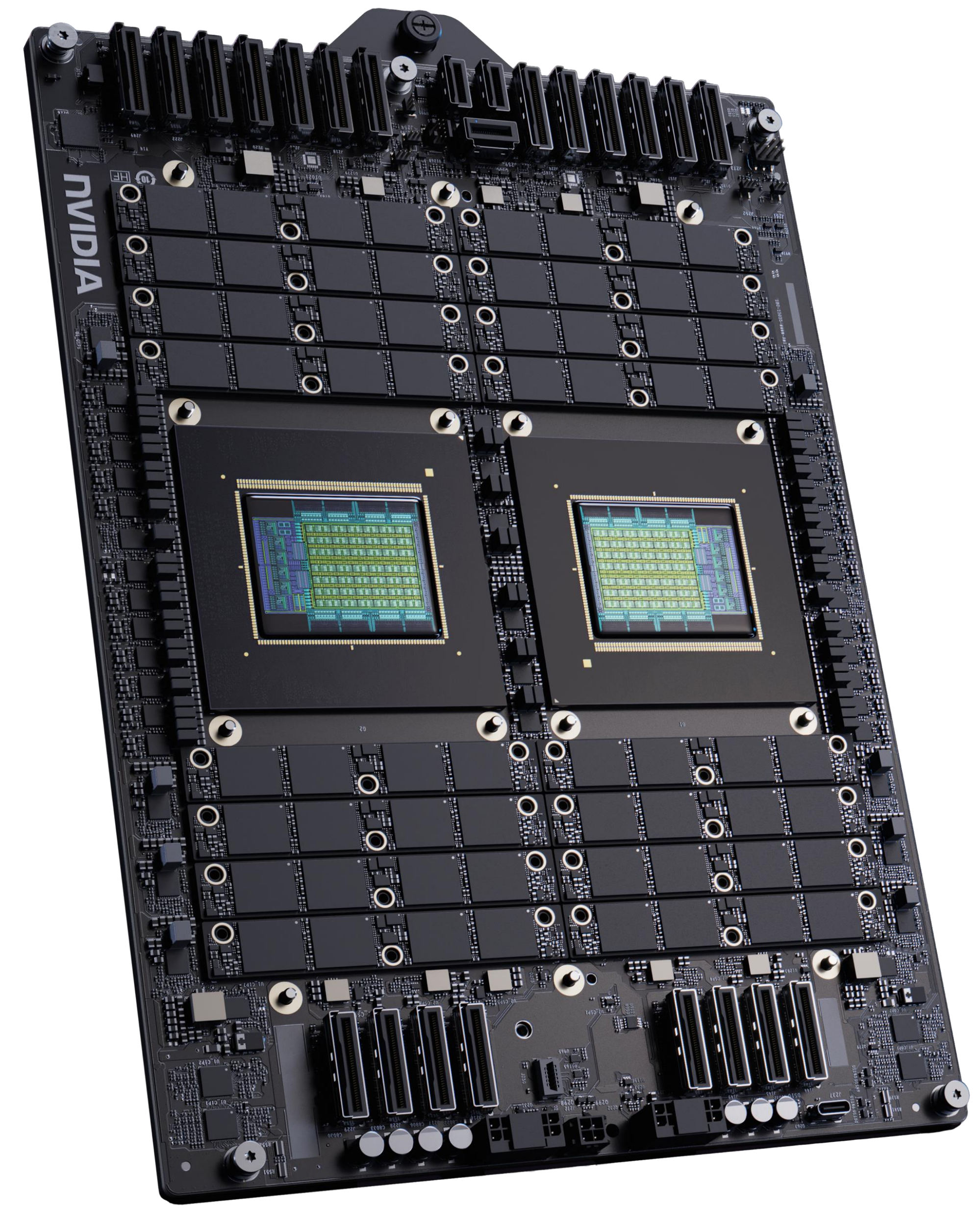
NVIDIA Vera, designed for agentic:

- High performance, power efficient cores
- Stable core performance across load (well provisioned for power and bandwidth)



Industry-First LPDDR5X Maximizes Memory Bandwidth Within Data Center Power Limits

Data center-class LPDDR5X implementation

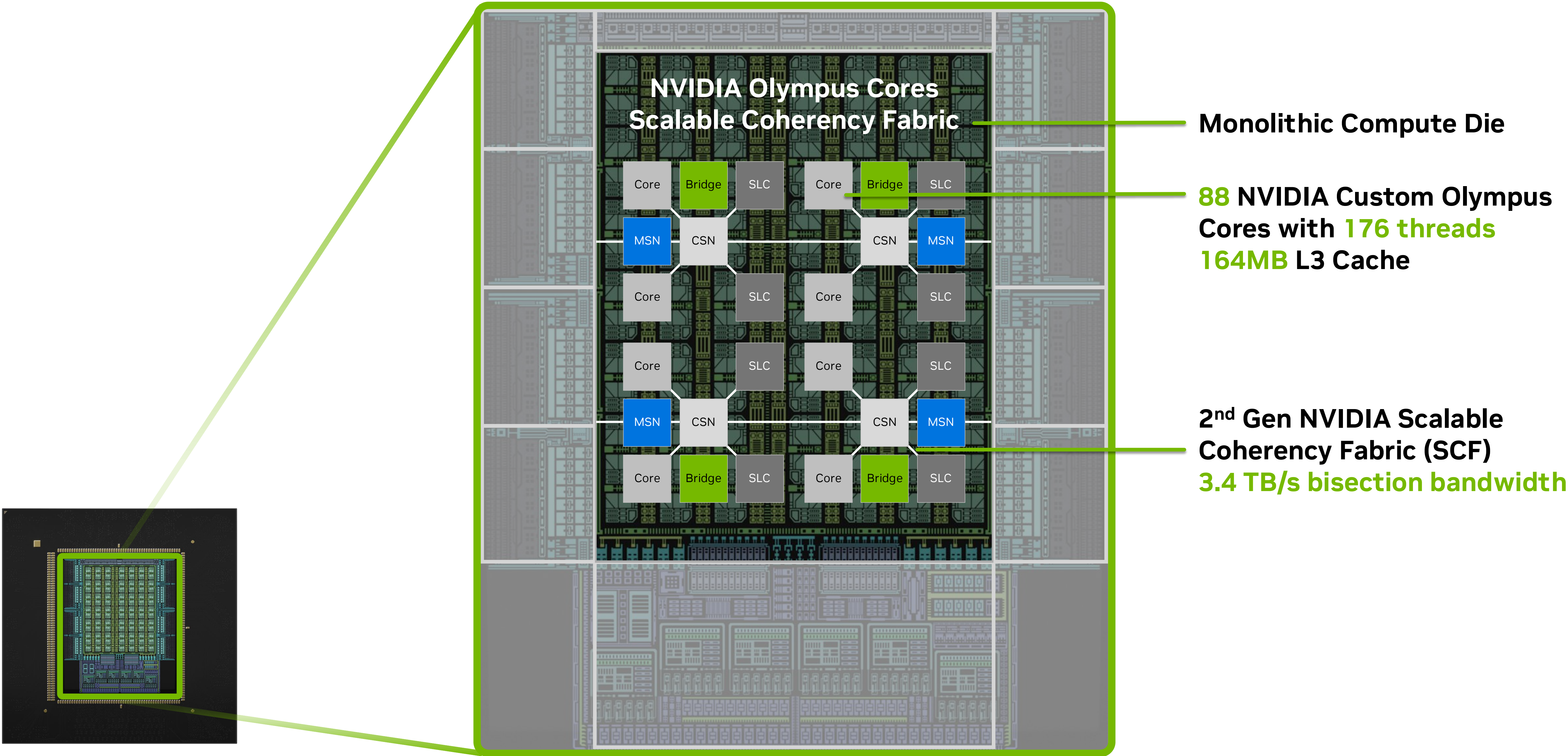


Differentiator 3: Built to Process Massive Data



Scalable Coherency Eliminates Data Ingestion Bottleneck

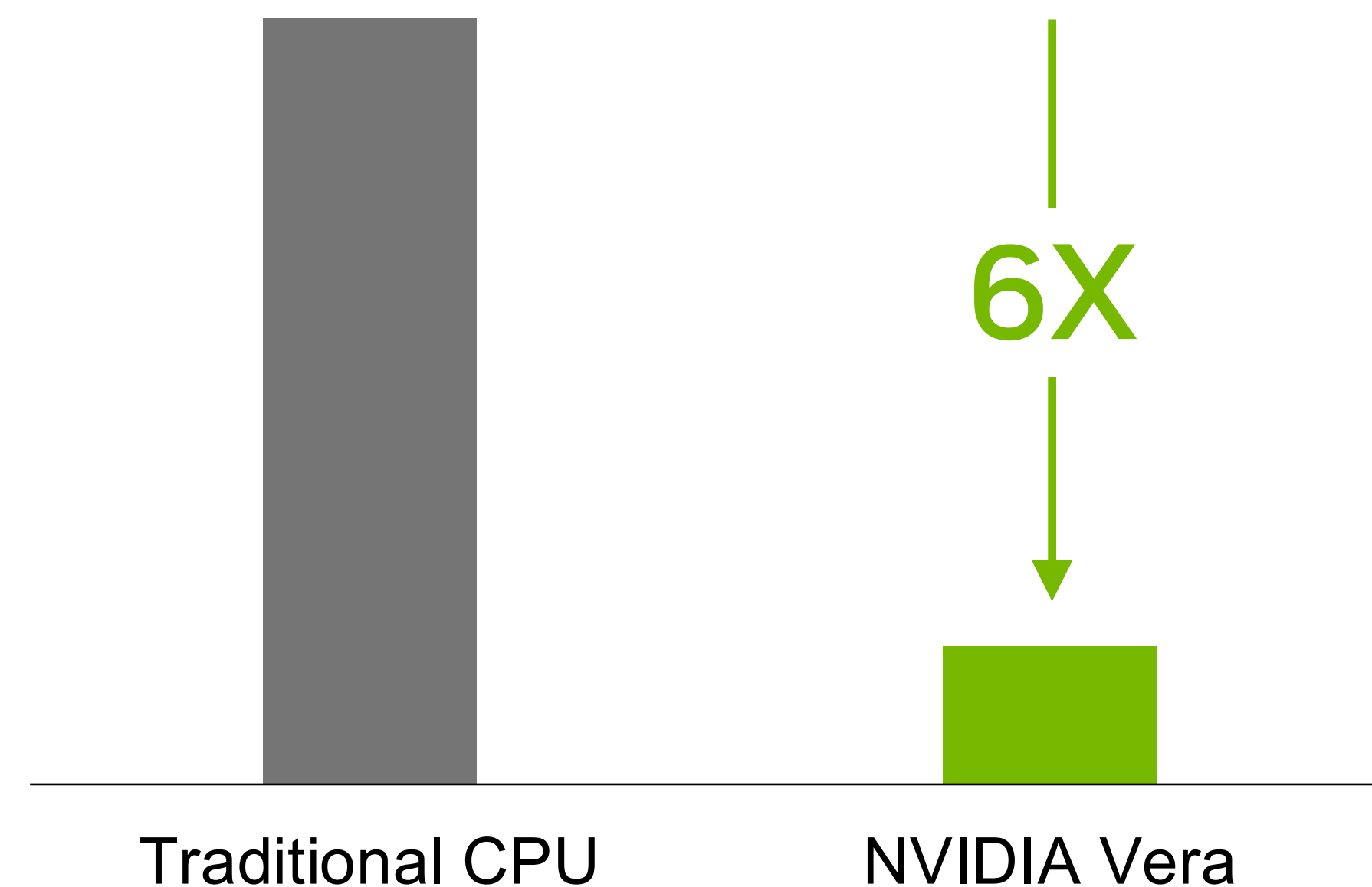
Vera designed to support core collaboration



Fabric mesh simplified for illustrative purposes

Vera Delivers up to 6X more Data Processing Performance

NYSE



“In collaboration with Redpanda and HPE, using NVIDIA Vera CPUs, we will be scaling our capacity, while further optimizing latency to power a high-performance, resilient and AI-ready market infrastructure.”

Lynn Martin – President of NYSE Group

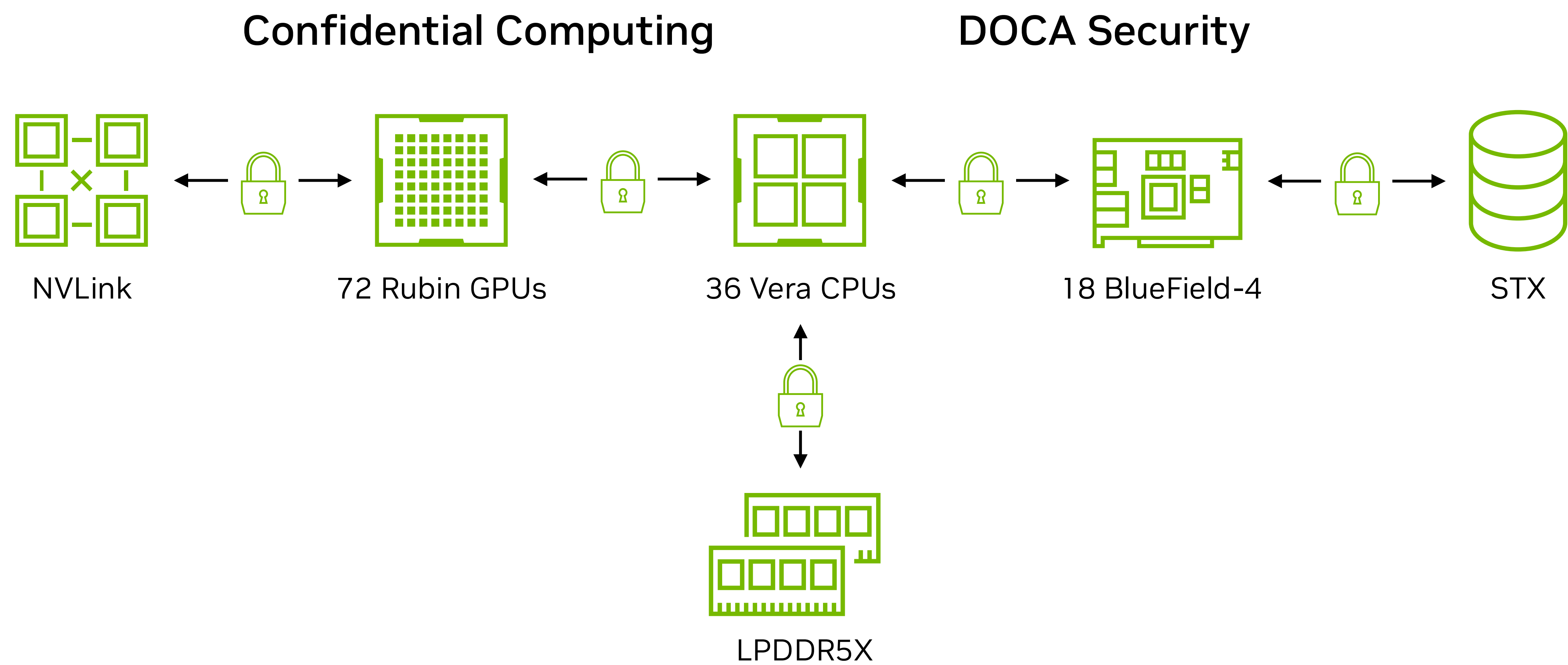
Differentiator 4: Completely Protected



Confidential Computing and DOCA

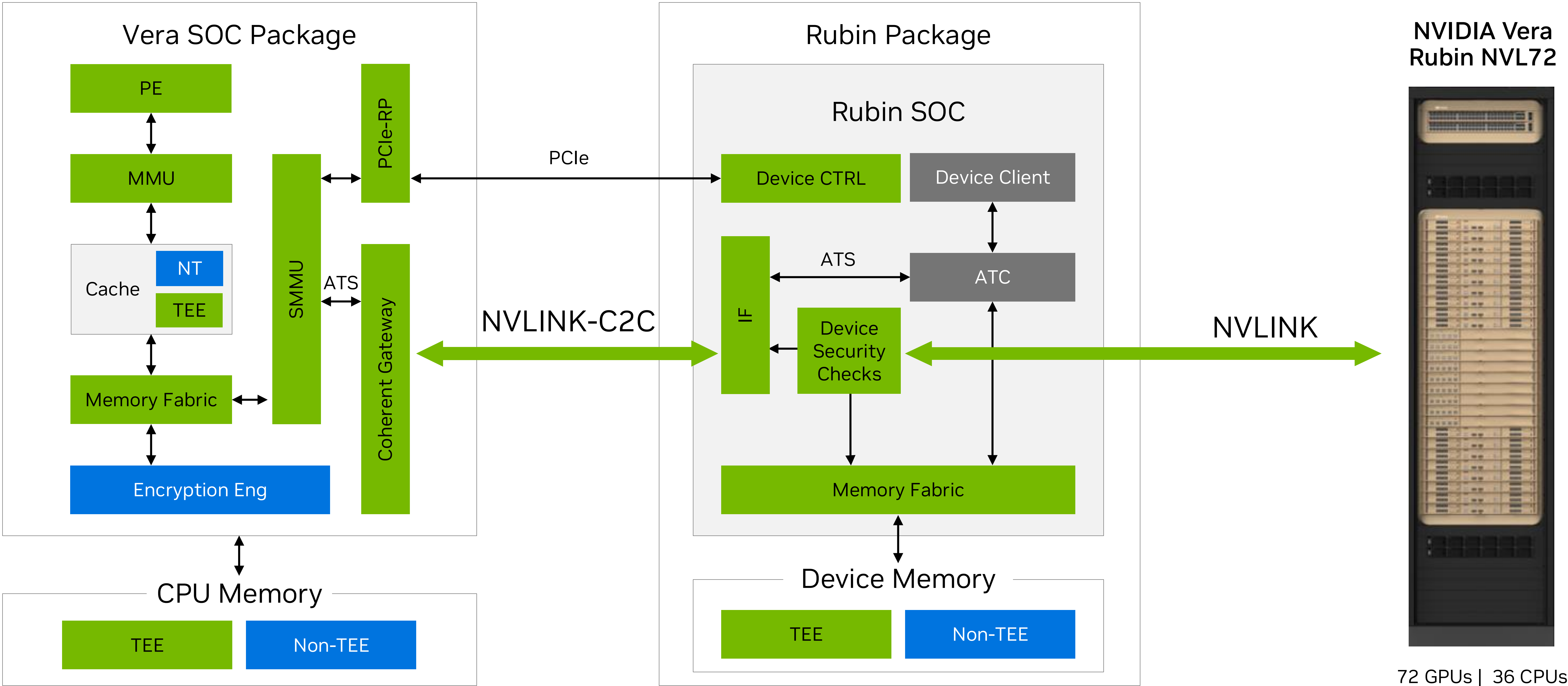
Secure the AI Factory End-to-End

Vera Rubin platform security



Confidential Compute Secures the Full Rack

Deep co-design at work

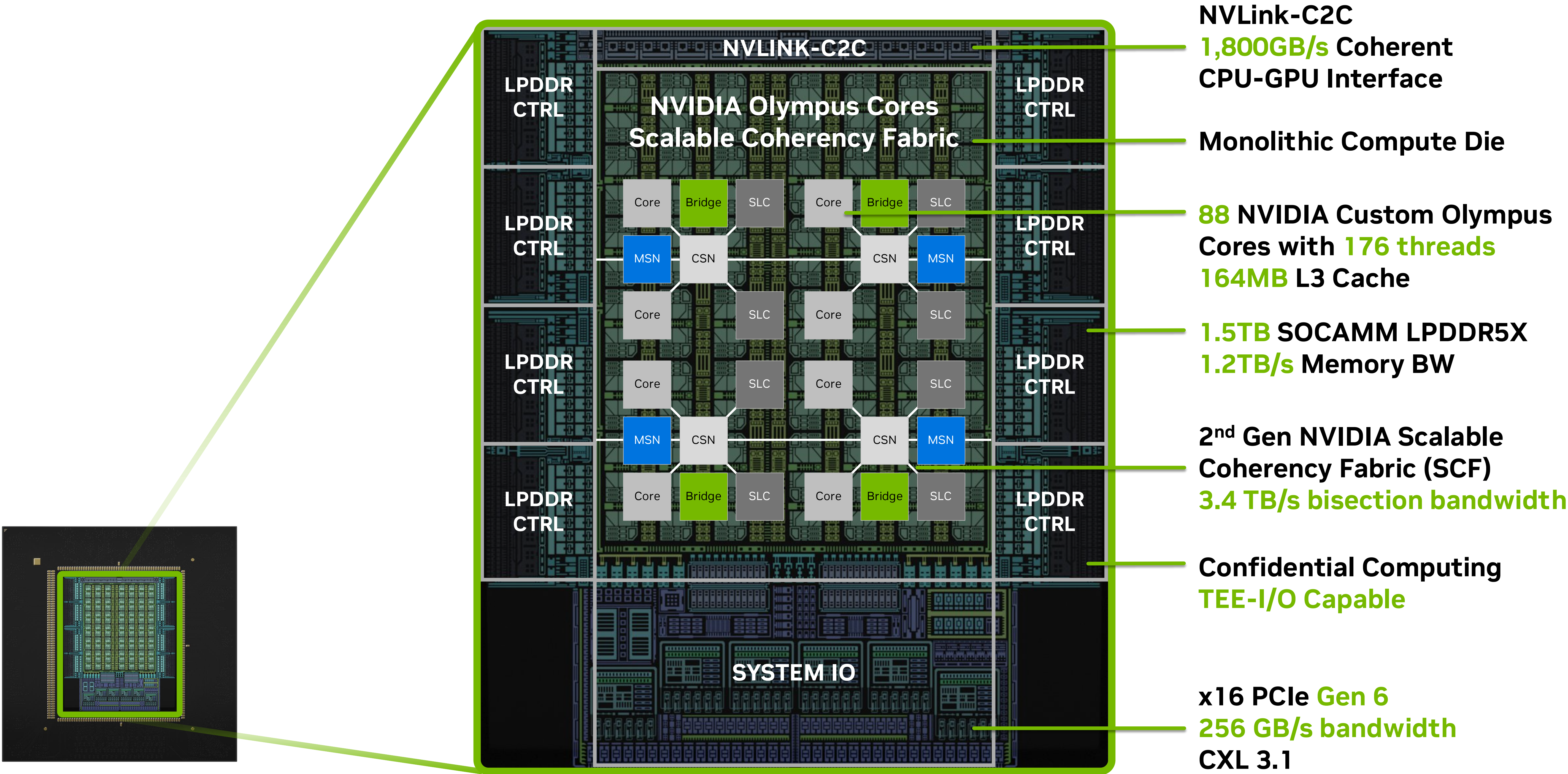


Differentiator 5: Massive I/O



Vera's Massive I/O Keeps GPUs and Memory Investments Fully Utilized

4.3 TB/s aggregate off die bandwidth keeping the AI factory flowing



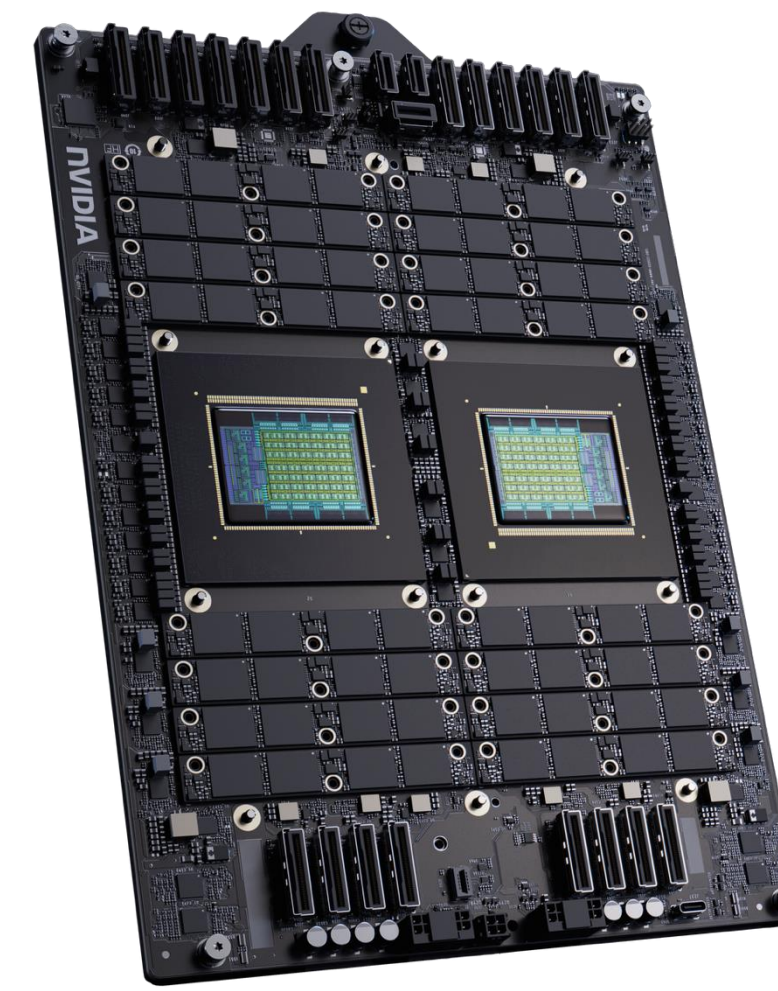
Fabric mesh simplified for illustrative purposes

Closing



Vera Rubin Co-Designs Every Layer of the AI Factory for Agentic AI

NVIDIA Vera CPU



- NVIDIA's second-generation server SOC
- Designed for agentic workloads
- NVIDIA Olympus cores
- Rack scale confidential compute

Thanks to the Vera CPU Team

NVIDIA Vera CPU



Standard Enterprise Air
Cooled 2S CPU Server

NVIDIA Vera Liquid
Cooled Rack



256 Veras
22,528 Olympus Cores

Vera Rubin
NVL72



72 GPUs
36 CPUs

